Vahan Product Analytics Case Study

Cohort Performance, Funnel Analysis & FT Prediction

Metric	Result
Uploaded leads	18,198
FT after upload	54
Overall FT conversion	0.297%
Top cohort	Single Referral > 7 days- 24th Jul — 0.933%
Final model	Funnel-stage Logistic Regression
Holdout ROC-AUC	0.7815
Holdout PR-AUC	0.0130

1. Executive Summary

This analysis evaluates lead-source cohort performance, funnel progression, and the ability to predict FT after upload. The dataset contains 18,198 leads, of which only 54 resulted in FT after upload, giving an overall FT conversion rate of 0.297%. This extreme class imbalance makes precision-recall analysis and business-oriented ranking more appropriate than accuracy alone.

The strongest historical cohorts were Single Referral > 7 days- 24th Jul (0.933%), Khanna- 2W 26th Jul (0.906%), and PreOb-Ob Fees Paid 29th Jul (set 1) (0.472%). The first two cohorts achieved more than three times the overall FT conversion rate.

2. Question 1 — Best Cohorts

FT after upload conversion was used as the primary cohort metric because FT is the final business outcome and the rate normalizes for different cohort sizes. Cohorts with fewer than 100 uploaded leads were excluded from the ranking because their rates are too unstable to support meaningful comparison.

Rank	Cohort	Leads	FT	FT conversion
1	Single Referral > 7 days- 24th Jul	1,500	14	0.933%
2	Khanna- 2W 26th Jul	1,546	14	0.906%
3	PreOb-Ob Fees Paid 29th Jul (set 1)	1,483	7	0.472%

The first two cohorts perform at approximately 3.15× and 3.05× the overall FT rate, respectively. This suggests meaningful differences in cohort quality rather than lead volume alone determining FT output.

3. Question 2 — SQL / Funnel Aggregation

The lead-level data was aggregated by lead_source using DuckDB. Counts were summed first and conversion rates were calculated from those aggregate counts, avoiding distortion from row-level NULLs or zero denominators.

A notable contrast is OLX - Ashwin - 2W - 17 Jul: it contains 5,182 leads but only 4 FT outcomes (0.077%). Single Referral > 7 days- 24th Jul generates 14 FT outcomes from 1,500 leads (0.933%). Therefore, acquisition volume should be evaluated alongside downstream FT efficiency.

The PreOb cohorts show strong Connect → Interested rates (15.85% and 16.60%) but lower final FT conversion (0.472% and 0.449%), demonstrating that strength at an intermediate funnel stage does not automatically translate into FT.

4. Question 3 — FT Prediction Model

The final model is a Logistic Regression using lead_source and funnel_stage. Funnel stage represents the deepest observed progression: 0 = not attempted, 1 = attempted, 2 = connected, 3 = tag filled, and 4 = interested. Downstream FT/OB outcomes were excluded from predictors to avoid target leakage.

Evaluation	ROC-AUC	PR-AUC
Logistic Regression — initial CV	0.7769	0.00965
Random Forest — initial CV	0.7702	0.01016
Extra Trees — initial CV	0.7604	0.00984
Final Funnel-stage Logistic Regression — holdout	0.7815	0.0130
Final model — cohort validation	0.6471	0.0064

5. Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	2,483	1,146
Actual 1	4	7

At the 0.5 threshold, the model identifies 7 of 11 FT leads, giving 63.64% recall. Precision for the FT class is only 0.61%, reflecting the extreme rarity of FT and the large number of false positives. Accordingly, accuracy (68.41%) should not be treated as the main business metric.

6. Factors Associated with FT

The strongest model associations are lead-source indicators and funnel stage. Khanna- 2W 26th Jul and Single Referral > 7 days- 24th Jul have the largest positive coefficients, consistent with their high observed cohort FT rates. Funnel stage has a positive coefficient (0.606), indicating that deeper observed funnel progression is associated with higher FT propensity in the model.

These are predictive associations, not causal effects. Because class weighting was used to address the rare FT class, coefficient odds ratios should not be interpreted as direct real-world probability multipliers.

7. Generalization to New Cohorts

When entire lead-source cohorts are kept together during validation, performance falls from holdout ROC-AUC 0.7815 to 0.6471, and PR-AUC falls from 0.0130 to 0.0064. This indicates that some predictive signal is cohort-specific and that performance on completely new acquisition cohorts may be materially lower than random holdout performance.

8. Business Recommendations

1. Prioritize and investigate high-performing acquisition cohorts, particularly Single Referral and Khanna-2W. 2. Evaluate lead sources using downstream FT efficiency rather than lead volume alone. 3. Use funnel progression as a prioritization signal for follow-up. 4. Use the ML model primarily for ranking/prioritization rather than strict FT classification. 5. Monitor model performance on new cohorts and retrain as acquisition sources change. 6. Increase the number of observed FT outcomes to improve statistical stability and model reliability.

9. Limitations

Only 54 FT outcomes are present in 18,198 leads, creating severe class imbalance and wide uncertainty around cohort-level rates. The analysis is observational and therefore does not establish causality. The decline in cohort-level validation performance also indicates that generalization to unseen sources is limited.

Supporting notebook: Vahan_Product_Analytics_Case_Study.ipynb

Supporting workbook: Vahan_Case_Study.xlsx